

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 9 — Plant Reproduction

ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. Asexual reproduction in which a cell splits into two equal parts: C — Binary fission 1 mark

Binary fission is the splitting of one cell into two equal daughter cells after DNA replication. It is common in bacteria. **Budding** produces an unequal outgrowth; **spore formation** involves production of dispersal units called spores; **fragmentation** is not listed as a main mode here.

Q2. Tiny outgrowth on yeast parent cell that detaches: B — Budding 1 mark

In **budding**, a small outgrowth (bud) forms on the parent yeast cell. The nucleus splits by mitosis, one nucleus enters the bud, the bud enlarges and eventually detaches. Binary fission splits the cell equally, not unequally.

Q3. Parthenogenesis in flowering plants is also known as: A — Apomixis 1 mark

Apomixis is the term used specifically for parthenogenesis in flowering plants. It is the asexual formation of a seed from the maternal tissues of the ovule, avoiding meiosis and fertilization. Pollination and fertilization are separate events.

Q4. Compact, thickened underground stem enveloped by thick succulent leaves: C — Bulb 1 mark

A **bulb** (e.g. onion, tulip) is a compact underground stem surrounded by thick, fleshy (succulent) leaves that store nutrients. A **corm** (e.g. colocasia) stores food in the stem itself, not in surrounding leaves. A **rhizome** is a horizontal underground stem.

Q5. Bryophyllum produces new plantlets along its: C — Leaves 1 mark

In **Bryophyllum**, tiny plantlets form along the notched edges of its leaves. When they detach and fall on soil, they grow into independent plants. This is vegetative propagation by leaves, unique to this plant.

Q6. Upper part of a graft carrying desired characteristics: B — Scion 1 mark

The **scion** is the upper part of the graft; it carries the desired traits such as fruit quality, flower colour, or disease resistance. The **rootstock** is the lower part, providing the root system and stem support. An **explant** is used in tissue culture, not grafting.

Q7. First piece of parent plant used in tissue culture: C — Explant 1 mark

The **explant** is the initial cell or small tissue piece taken from the parent plant and placed on a nutrient medium. It grows into an undifferentiated mass called a **callus**, which is then induced to form a plantlet. The term "plantlet" refers to the fully formed young plant at the end.

Q8. Two whorls called necessary whorls (directly involved in reproduction): B — Androecium and gynoecium 1 mark

The **androecium** (male, produces pollen) and **gynoecium** (female, contains the ovule/egg) are the necessary whorls because they directly participate in reproduction. **Calyx** and **corolla** are accessory whorls that protect or attract pollinators.

Q9. Haploid pollen grains produced by meiosis in the anther: B — Microspores 1 mark

Microspores are produced by meiosis in the four pollen sacs (microsporangia) of the anther. They become pollen grains. **Megaspores** are the female equivalent, produced in the ovule. **Synergids** are cells found inside the embryo sac.

Q10. Pollination between flowers of two separate plants: C — Cross-pollination 1 mark

Cross-pollination occurs when pollen from one plant reaches the stigma of a flower on a different plant of the same species. It ensures more genetic variability in offspring. **Self-pollination** occurs within the same plant and produces genetically uniform offspring.

Q11. Second sperm fuses with polar nuclei to form: C — Triploid endosperm nucleus 1 mark

In **double fertilization**, one sperm (1N) fuses with the egg (1N) to form a diploid (2N) zygote. The second sperm (1N) fuses with the two polar nuclei (1N + 1N) to form a **triploid (3N) endosperm nucleus**. The endosperm later stores food for the developing embryo.

Q12. Part that stretches and pulls cotyledons above ground in epigeal germination: C — Hypocotyl 1 mark

In **epigeal germination**, the **hypocotyl** (stem below cotyledon attachment) elongates and forms a hook, pulling the cotyledons above the soil. Examples: bean, cotton, papaya. In hypogeal germination it is the **epicotyl** that elongates while cotyledons stay underground.

SECTION B — Short Questions (11 × 3 = 33 Marks)

2(i) Define asexual reproduction. State three characteristics that make it different from sexual reproduction.

3 marks

Asexual reproduction is a method in which only one parent is involved and offspring are generated that are genetically identical to the parent (clones). It does not require gamete formation or meiosis. **1 mark**

Three distinguishing characteristics:

- Only one parent is needed; sexual reproduction requires two parents. **1 mark**
- No gametes or fertilization are involved; sexual reproduction requires gamete fusion.
- Offspring are genetically identical (no variation); sexual reproduction produces genetically diverse offspring. **1 mark**

OR

Describe binary fission in bacteria. **3 marks**

In **binary fission**, the bacterium first replicates its DNA so two identical sets are formed. **1 mark** The two DNA sets move to opposite ends of the cell, and the middle of the cell folds inward. **1 mark** A cross-wall (septum) then forms completely, dividing the original cell into two equal daughter cells, each with one copy of the DNA. **1 mark**

2(ii) Explain budding in yeast. What happens to the nucleus during bud formation? **3 marks**

In **budding**, a small outgrowth called a bud emerges from the parent yeast cell. **1 mark** Inside the parent cell, the nucleus splits by **mitosis** and one of the new nuclei moves into the bud. **1 mark** The bud grows larger, becoming more like the parent, and eventually detaches as a new independent cell. Sometimes multiple buds form on the same parent simultaneously. **1 mark**

OR

What is spore formation? How does Rhizopus produce and release spores? **3 marks**

Spore formation is an asexual method in which small, unicellular, thick-walled, non-motile reproductive bodies called **spores** are produced. **1 mark** In Rhizopus, body cells form protective bags called **sporangia** on the tips of stalks called **sporangiophores**. Cells inside divide to produce many spores, each with a tough cyst wall. **1 mark** When the sporangia are mature they burst open, releasing spores. Under favourable conditions, each spore germinates into a new Rhizopus.

1 mark

2(iii) What is vegetative propagation? Distinguish natural from artificial with one example each. **3 marks**

Vegetative propagation is asexual reproduction in plants in which new individuals are produced from vegetative structures (roots, stems, leaves) without seeds. **1 mark**

- **Natural vegetative propagation:** occurs without human involvement using the plant's own structures (e.g. strawberry runners/stolons). **1 mark**
- **Artificial vegetative propagation:** humans purposely use a vegetative propagule to grow plants with desired traits (e.g. stem cuttings of roses). **1 mark**

OR

Define parthenogenesis. Why is it asexual? What is it called in flowering plants? **3 marks**

Parthenogenesis is the development of an offspring directly from an unfertilized egg. **1 mark** It is considered asexual because no fusion of genetic material from two parents occurs; the offspring are produced without fertilization. **1 mark** In flowering plants it is called **apomixis** and involves the asexual formation of a seed from the maternal tissues of the ovule, bypassing meiosis and fertilization. **1 mark**

2(iv) Describe any three methods of natural vegetative propagation with one plant example each. **3 marks**

- **Bulb:** compact, thickened underground stem surrounded by fleshy leaves storing nutrients; adventitious roots and shoots arise from it. Example: onion. **1 mark**
- **Rhizome:** flat, horizontal underground stem with distinct nodes and internodes; buds on the upper side grow into shoots. Example: ginger. **1 mark**
- **Stolon/Runner:** long horizontal above-ground stem that grows outward; at nodes, new shoots and roots develop into independent plants. Example: strawberry. **1 mark**

OR

Differentiate bulb and rhizome with one example each. How does each give rise to a new plant? **3 marks**

Feature	Bulb	Rhizome
Orientation	Vertically growing, compact	Horizontal, flat underground stem
Food storage	In thick, succulent (fleshy) leaves	In the stem tissue itself
New plant formation	Adventitious roots below, shoots from buds above	Buds on upper nodes grow into shoots; lower side forms roots
Example	Onion, tulip, lily	Ginger, fern, water lily

1 mark per two correct rows / example pair; full table = 3 marks.

2(v) Explain stem cuttings. What role does auxin play? 3 marks

Stem cuttings are small pieces of stem, each with at least one node, cut from the parent plant. **1 mark** The cutting is placed in water or damp soil; roots grow from the bottom of the stem while the top produces leaves. **1 mark** If the cutting fails to root, the cut ends are treated with the plant hormone **auxin**, which stimulates root initiation and promotes growth. The top may also be covered with plastic or glass to reduce water loss. **1 mark**

OR

Describe grafting. Name the two parts and explain the role of each. 3 marks

Grafting joins two plant parts to create a single new plant and is used to reproduce desirable varieties. **1 mark**

- **Scion:** the upper part carrying desired characteristics (fruit type, flower colour). It contributes the desired traits to the grafted plant. **1 mark**
- **Rootstock:** the lower part providing the established root system and stem support. Precise cuts maximize contact between scion and rootstock; grafting tape holds them together until they fuse and the plant begins to grow. **1 mark**

2(vi) What is tissue culture? Define explant and describe what happens in the nutrient medium. 3 marks

Tissue culture (cloning / micropropagation) is a method of producing genetically identical individuals by growing a single plant cell or piece of tissue on an artificial nutrient medium under aseptic conditions. **1 mark** The first cell or tiny tissue piece taken from the parent plant is called the **explant**. **1 mark** The explant is placed on a nutrient medium and grows into an undifferentiated mass of cells called a **callus**. The callus is then treated with substances that cause it to produce shoots and roots, eventually developing into a complete **plantlet**, which is then transferred to natural soil. **1 mark**

OR

Differentiate artificial and natural vegetative propagation. Why is artificial preferred commercially? 3 marks

Feature	Natural Vegetative Propagation	Artificial Vegetative Propagation
Human involvement	None; uses plant's own structures	Humans purposely use a vegetative propagule
Methods	Bulbs, rhizomes, runners, suckers	Stem cuttings, grafting, tissue culture
Trait control	No selection of specific traits	Desired traits (taste, disease-resistance) preserved

Artificial propagation is preferred commercially because it preserves desirable qualities (taste, colour, disease-resistance) across large numbers of plants and can be performed at scale in controlled conditions. **1 mark**

2(vii) Name the four floral whorls. Which are necessary, which are accessory? 3 marks

A typical flower possesses four whorls arranged on the thalamus: **1 mark**

- **Calyx** (sepals) and **Corolla** (petals): the outer two whorls. They protect the inner flower and attract pollinators but do not directly participate in reproduction. They are called **accessory whorls**. **1 mark**
- **Androecium** (stamens) and **Gynoecium** (carpels/pistil): the inner two whorls. They directly carry out reproduction (produce pollen and house the egg) and are called **necessary whorls**. **1 mark**

OR

Describe the structure of a stamen. How are microspores (pollen grains) produced? 3 marks

Each **stamen** consists of a long stalk called the **filament** and a bilobed swollen tip called the **anther**. **1 mark** In the anther lobes, there are four pollen sacs called **microsporangia**. **1 mark** Inside each microsporangium, **microspore mother cells** undergo meiosis to produce haploid (1N) pollen grains (**microspores**). When the pollen sacs are mature, the anther wall bursts and the microspores are liberated. **1 mark**

2(viii) Define pollination. Distinguish self-pollination from cross-pollination. Which promotes genetic variability? 3 marks

Pollination is the transfer of microspores (pollen grains) from the anther of a stamen to the stigma of a carpel by wind, insects, or other animals. **1 mark**

- **Self-pollination:** pollen is transferred within the same flower or between two flowers of the same plant. Offspring are genetically similar. **1 mark**
- **Cross-pollination:** pollen is transferred between flowers of two separate plants. This ensures more **genetic variability** in coming generations because gametes from two genetically different parents combine. **1 mark**

OR

Compare insect-pollinated and wind-pollinated flowers under three features. 3 marks

Feature	Insect-pollinated	Wind-pollinated
Petal colour	Brightly coloured (attracts insects)	Green or dull colour (no need to attract)
Pollen grains	Small number produced; heavy and sticky (clings to insect)	Large number produced; light with smooth surface (blown by wind)
Stigma shape	Pinhead-shaped with no branches	Feathery branches to catch airborne pollen

1 mark per correctly compared feature. Total: 3 marks.

2(ix) What is the female gametophyte (embryo sac)? Name the cell types inside it. 3 marks

The female gametophyte (embryo sac) is the mature structure inside the ovule of an angiosperm that houses the female gametes. It develops from the functional megaspore. 1 mark The embryo sac contains eight nuclei organized into the following cell types: 1 mark

- **Egg cell** (1N): the female gamete, located in the center at one end of the ovule.
 - **Synergids** (1N, two cells): flank the egg on both sides; assist in guiding the pollen tube.
 - **Antipodal cells** (1N, three cells): at the opposite end of the ovule; control the fertilization process.
 - **Polar nuclei** (two 1N nuclei): at the center; fuse with the second sperm to form the triploid endosperm nucleus.
- 1 mark

OR

Describe the development of the male gametophyte after pollination. Name the two nuclei. What does the pollen tube do? 3 marks

After pollination, the microspore (pollen grain) germinates on the surface of the stigma. 1 mark The microspore nucleus divides by mitosis to form two nuclei: the tube nucleus and the generative nucleus. The generative nucleus further divides to form two sperm nuclei. 1 mark The tube nucleus controls the formation of the pollen tube, which grows down through the style and penetrates the ovule. This whole structure (microspore body + pollen tube + tube nucleus + two sperm nuclei) is the male gametophyte. 1 mark

2(x) Name the three parts of an angiosperm seed. Briefly explain the function of each. 3 marks

- **Embryo:** develops from the zygote (union of egg and sperm). It is an immature plant consisting of the radicle, plumule, and cotyledons. It will grow into the new plant upon germination. 1 mark
- **Endosperm:** storage tissue derived from the triploid endosperm nucleus. Rich in starch, oil, and protein; provides nutrients to the growing seedling. 1 mark
- **Seed coat (testa):** the protective outer covering that develops from the integuments surrounding the ovule. It protects the embryo from drying, mechanical damage, and pathogens. 1 mark

OR

Explain seed and fruit formation after fertilization. What does the ovule become? What does the ovary wall become? 3 marks

After double fertilization, the zygote (2N) develops into the embryo and the triploid endosperm nucleus (3N) develops into the endosperm tissue. 1 mark The ovule is transformed into the seed (defined as a "ripened ovule"). 1 mark The surrounding wall of the ovary is transformed into the fruit (defined as a "ripened ovary"). The fruit protects the seed and often aids in its dispersal. 1 mark

2(xi) State three external conditions for seed germination. Explain the role of water. 3 marks

Three external conditions required for seed germination: 1 mark

- **Water (moisture):** seeds absorb water, which causes them to swell and break open the seed coat. Water activates the stored food reserves and provides a medium for metabolic reactions to start. 1 mark
- **Oxygen:** needed by the embryo cells for cellular respiration, which provides energy for the growing seedling.
- **Suitable temperature:** most seeds germinate optimally at 25–30°C; temperature controls enzyme activity during germination. 1 mark

OR

Differentiate epigeal and hypogeal germination. Give one example each and name the part that elongates. 3 marks

Feature	Epigeal Germination	Hypogeal Germination
Part that elongates	Hypocotyl (stem below cotyledon attachment)	Epicotyl (stem above cotyledon attachment)
Cotyledon position	Pulled above ground; cotyledons are exposed to light	Cotyledons remain underground; only shoot emerges
Example	Bean, cotton, papaya	Pea, maize, coconut

1 mark per pair of correctly contrasted features. Total: 3 marks.

SECTION C — Long Questions (4 × 5 = 20 Marks)

3(i) Compare sexual and asexual reproduction in plants under five points. Which is more beneficial for changing environments? **5 marks**

Feature	Asexual Reproduction	Sexual Reproduction
Number of parents	One parent only	Two parents (male and female)
Gamete formation	Not required; no meiosis	Required; meiosis produces gametes
Genetic variation	Offspring are genetically identical (clones)	Offspring are genetically diverse
Speed	Rapid; can produce many offspring quickly	Slower; complex process of gamete fusion
Suitability	Suitable for stable, favourable environments	Suitable for changing or unfavourable environments

1 mark per two correctly compared rows; table correct = 4 marks; conclusion = 1 mark.

Conclusion: Sexual reproduction is more beneficial for adapting to changing environments because it generates genetic variability in offspring. This variation means that some individuals may carry traits better suited to new conditions, increasing the chance that at least some offspring survive environmental changes. **1 mark**

OR

Describe the methods of natural vegetative propagation. Cover all seven types with one example each. **5 marks**

Natural vegetative propagation is the process by which plants reproduce asexually using their own vegetative structures without human involvement. Seven methods:

- **Bulb** (e.g. onion): compact underground stem enveloped by fleshy nutrient-storing leaves; adventitious roots and buds produce new shoots. **0.5 mark**
- **Corm** (e.g. colocasia/dasheen): compact underground stem that stores food in the stem itself; buds at the top give rise to new shoots. **0.5 mark**
- **Rhizome** (e.g. ginger): horizontal underground stem with nodes and internodes; upper buds grow into shoots, lower side produces adventitious roots. **0.5 mark**
- **Stem tuber** (e.g. potato): swollen tips of underground stems with "eyes" (axillary buds); new shoots grow from the buds under suitable conditions. **0.5 mark**
- **Stolon/Runner** (e.g. strawberry): horizontal above-ground stem; at nodes, shoots and roots develop into independent plants; the runner eventually dries up. **0.5 mark**
- **Suckers** (e.g. lawn grass): small shoots that emerge from the underground root system or stem base; they develop their own roots and become self-sustaining. **0.5 mark**
- **Vegetative propagation by leaves** (e.g. Bryophyllum): plantlets form along the notched edges of leaves; they detach, fall on soil, and grow into independent plants. **1 mark**

3(ii) Describe the structure of a typical flower. Name and explain the function of each of the four whorls. Label key parts of the gynoecium and androecium. (5 marks)

A **flower** is the reproductive organ of an angiosperm. It consists of a stalk called the **pedicel** topped by the **thalamus**, from which four whorls of floral leaves arise. (1 mark)

- **Calyx** (1st whorl, outermost): made up of green leaf-like **sepals**. Function: protects the inner flower parts before it opens; accessory whorl. (1 mark)
- **Corolla** (2nd whorl): made up of variously coloured **petals**. Function: attracts pollinators (insects) through colour and scent; accessory whorl. (1 mark)
- **Androecium** (3rd whorl): the male reproductive part made up of **stamens**. Each stamen = **filament** (stalk) + **anther** (tip). The anther contains four pollen sacs (microsporangia) that produce pollen grains by meiosis; necessary whorl. (1 mark)
- **Gynoecium** (4th whorl, innermost): the female reproductive part made up of one or more **carpels (pistils)**. Each carpel = **stigma** (sticky tip to catch pollen) + **style** (tube) + **ovary** (base, contains ovules). The ovule houses the megaspore that develops into the egg; necessary whorl. (1 mark)

OR

Explain plant tissue culture (cloning/micropropagation). Define explant, describe callus formation, explain how a plantlet develops. State two advantages. (5 marks)

Tissue culture (also called micropropagation or cloning) is the process of growing a single plant cell or small piece of tissue on an artificial nutrient medium under aseptic (microbe-free) conditions to produce a genetically identical copy (clone) of the parent plant. (1 mark)

```
Parent plant
  |
  v
[Explant taken] -- small cell or tissue piece from parent
  |
  v
[Placed on nutrient medium under aseptic conditions]
  |
  v
[Callus forms] -- undifferentiated mass of cells
  |
  v
[Treated with growth substances]
  |
  v
[Plantlet develops -- shoots + roots]
  |
  v
[Transferred to natural soil]
```

- The **explant** is the first cell or tiny piece of parent plant tissue used to begin the culture. (1 mark)
- On the nutrient medium, the explant grows into a **callus** (undifferentiated mass). The callus is then treated with different substances to stimulate it to produce shoots and later roots, forming a **plantlet**. (1 mark)
- The plantlet is carefully moved to natural soil to continue growing. (1 mark)

Two advantages: (1) preserves desirable plant traits exactly (taste, disease-resistance, yield); (2) produces large numbers of genetically identical plants rapidly from a single parent. (1 mark)

3(iii) Explain double fertilization in angiosperms. Name both fusion events, identify ploidy of each product, and explain why it is called "double" fertilization. (5 marks)

Double fertilization is unique to angiosperms and involves two separate fusion events using two sperm nuclei delivered by one male gametophyte. (1 mark)

```
Pollen grain germinates on stigma
|
v
Pollen tube grows down the style
|
v
Pollen tube penetrates the ovule
|
+--> Sperm nucleus 1 (1N) + Egg cell (1N)
|      --> Zygote (2N / diploid)
|      --> Embryo
|
+--> Sperm nucleus 2 (1N) + Two polar nuclei (1N + 1N)
|      --> Endosperm nucleus (3N / triploid)
|      --> Endosperm tissue (food store)
```

- **First fusion:** one sperm nucleus (1N) fuses with the **egg cell** (1N) to form the **zygote** (2N / diploid). The zygote develops into the embryo. (1 mark)
- **Second fusion:** the other sperm nucleus (1N) fuses with the two **polar nuclei** (1N + 1N) to form the **triploid endosperm nucleus** (3N). This develops into the endosperm, which stores nutrients for the embryo. (2 marks)
- It is called **"double" fertilization** because one male gametophyte contributes two sperm nuclei, and both fuse with different nuclei in the embryo sac in two separate, simultaneous fertilization events. (1 mark)

OR

Describe the development of the female gametophyte (embryo sac). How does the functional megaspore divide? Name all structures in a mature embryo sac. (5 marks)

The female gametophyte develops inside the **ovule**. The ovule contains a **macrospore mother cell** (2N) that undergoes meiosis to produce four megaspores; three degenerate and one becomes the **functional megaspore**. (1 mark)

The nucleus of the functional megaspore divides by mitosis **three times** to produce eight nuclei. (1 mark)

- Three nuclei migrate to one end of the ovule: the central one becomes the **egg cell** (1N) and the two on either side are called **synergids** (1N each). (1 mark)
- Three other nuclei migrate to the opposite end and act as **antipodal cells** (1N each). They help control the fertilization process. (1 mark)
- The remaining two nuclei rest in the center; they are called **polar nuclei** (1N each). In some cases they fuse before fertilization to form the diploid **secondary nucleus**. (1 mark)

This whole structure (megaspore body + egg + two synergids + three antipodals + two polar nuclei or secondary nucleus) is the **female gametophyte or embryo sac**.

3(iv) Define seed germination. Describe the angiosperm seed structure. Explain three external conditions and distinguish epigeal from hypogeal germination with one example each. 5 marks

Seed germination is the process by which a seed absorbs water, swells, breaks open the seed coat, and the embryo starts growing into a seedling under suitable conditions. 1 mark

Structure of an angiosperm seed (three parts):

- Embryo: an immature plant consisting of the radicle (future root), plumule (future shoot), epicotyl, hypocotyl, and one or two cotyledons. 0.5 mark
- Endosperm: nutrient storage tissue (starch, oil, protein) that feeds the growing embryo and seedling. 0.5 mark
- Seed coat (testa): protective outer layer; has a hilum (point of attachment) and micropyle (water entry point). 0.5 mark

Three external conditions for germination:

- Water: seeds absorb water, swell, break the seed coat, and activate enzymes to digest stored food for the embryo. 0.5 mark
- Oxygen: embryo cells require aerobic respiration for energy to fuel growth. 0.25 mark
- Temperature (25–30°C optimal): controls the rate of enzyme activity; too cold slows germination, too hot denatures enzymes. 0.25 mark

Feature	Epigeal Germination	Hypogeal Germination
Elongating part	Hypocotyl forms a hook and elongates	Epicotyl forms a hook and elongates
Cotyledon position	Cotyledons pulled above the soil surface	Cotyledons remain underground
Example	Bean, cotton, papaya	Pea, maize, coconut

1 mark for the comparison table (both types correctly described with examples).

OR

Compare insect-pollinated and wind-pollinated flowers using a table covering six features. Explain why each adaptation is necessary. 5 marks

Feature	Insect-pollinated Flowers	Wind-pollinated Flowers
Size	Generally large	Generally small
Colour	Petals brightly coloured	Petals green or dull
Nectar	Produce nectar to reward insects	Do not produce nectar
Floral arrangement	Flowers face upward	Flowers hang down for easy shaking
Pollen grains	Small number; heavy and sticky (clings to insect body)	Large number; light with smooth surface (carried by wind)
Stigma	Pinhead-shaped with no branches	Feathery/branched to trap airborne pollen

0.5 mark per feature row correctly contrasted (6 rows = 3 marks).

Why adaptations are necessary:

- Insect flowers need bright petals, nectar, and sticky pollen to attract insects and ensure pollen adheres to the insect body for transfer. 1 mark
- Wind flowers need light pollen (to travel far), large quantities (many will be lost), and feathery stigmas (to intercept airborne grains) because wind dispersal is random and inefficient. 1 mark